

CS & IT ENGINEERING



Computer Network

IPv4 Header

Lecture No. - 01



By - Abhishek Sir



Recap of Previous Lecture



Topic

Supernetting

Topic

NAT



Topics to be Covered



Topic

IPv4 Header



ABOUT ME



Hello, I'm **Abhishek**

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Topic : Supernetting



Example 4 :- Consider following Network Addresses of networks.

10 . 90 . 0 . 0 / 16

10 . 90 . 64 . 0 / 18

10 . 90 . 192 . 0 / 18

Supernet Address : 10 . 90 . 0 . 0 / 16



Topic : Supernetting



Example 4 :-

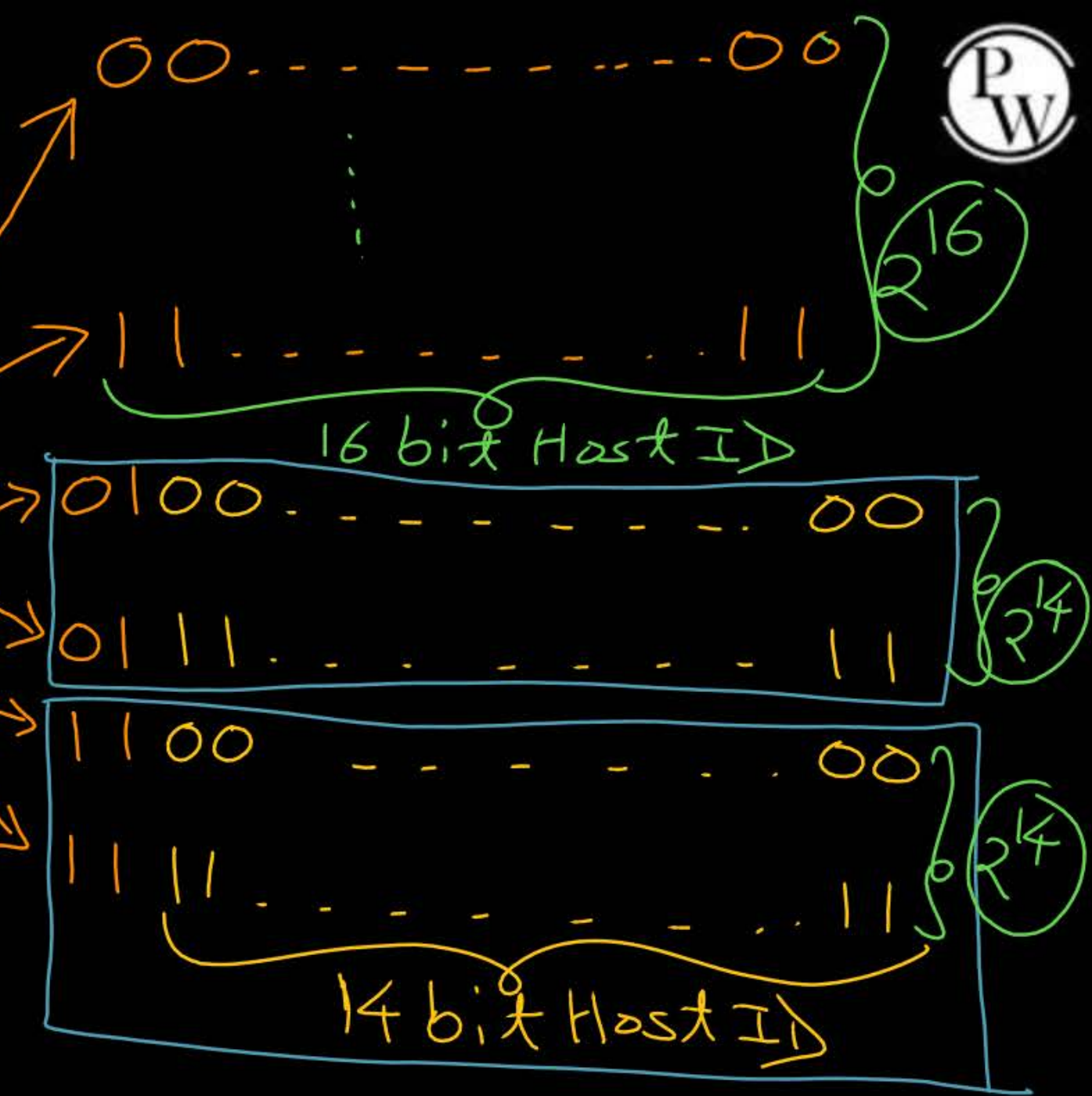
16 bit prefix
10.90.00000000.00000000 / 16
16 bit Host ID

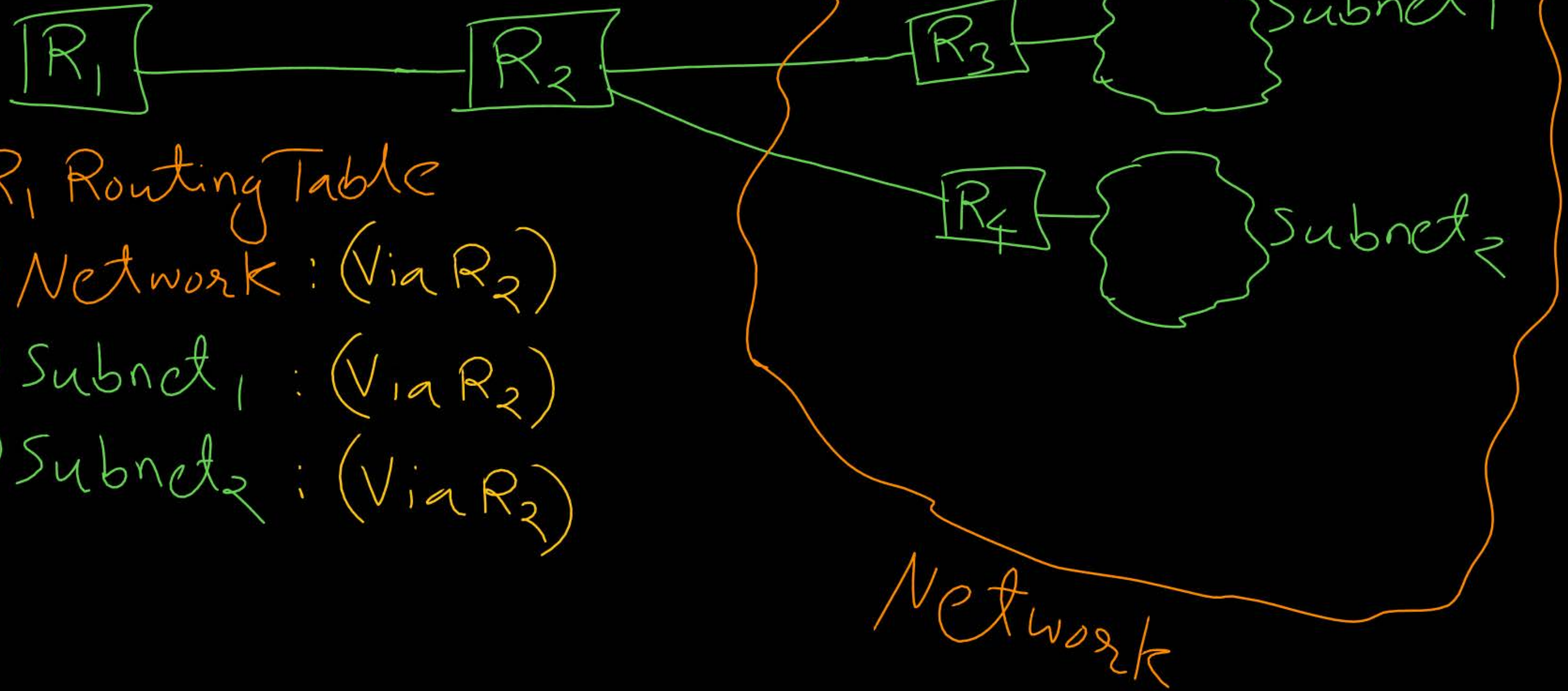
10.90.01000000.00000000 / 18
18 bit prefix

10.90.11000000.00000000 / 18
18 bit prefix
14 bit Host ID

Supernet Address : 10.90.0.0 /

10.90.00000000.00000000 /





R_1 Routing Table

- ① Network : (Via R_2)
- ② Subnet₁ : (Via R_2)
- ③ Subnet₂ : (Via R_2)



Topic : Network Layer



→ Internet Protocol (IP)

→ Two versions :

1. IPv4



2. IPv6





Topic : IPv4 Packet Header



IPv4 Packet / Datagram





Topic : IPv4 Packet Header



IPv4 Header
5 to 15 word
(20 to 60 byte)

0	3/4	7/8	15/16	31
VER	HLEN	Type of Services	Total Length	
Identification No.			DF MF	Fragmentation Offset
Time-to-Live		Protocol Type	Header Checksum	
Source IP Address (32 bits)				
Destination IP Address (32 bits)				
Options (Optional Header)				
—//— Payload —//—				

BASE
IPv4
Header
(5 word)
(20 byte)

0 to 10 word
(0 to 40 byte)



Topic : IPv4 Packet Header



- IPv4 Header represented in words
[Word of 32 bits (4 bytes)]
- Minimum (Base) Header Size = 5 Words (20 Bytes)
[Word size = 4 bytes]
- Variable Size IPv4 Header
[due to options (optional header)]



Topic : Version



- First four bits of IPv4 datagram
- “0100” : for IPv4 ✓
- “0110” : for IPv6



Topic : Header Length



→ Header Length [HLEN]

→ HLEN field is 4 bits long

→ Size of header in words (by default)
[Word of 4 bytes] ↑

$$\begin{aligned}\rightarrow \text{Header Length} &= [\text{HLEN}] \text{ Words} \\ &= [\text{HLEN} * 4] \text{ Bytes}\end{aligned}$$



Topic : Header Length

→ Minimum Header Size = 5 Words (20 Bytes)

$$5 \leq \text{HLEN} \leq 15$$

→ Maximum Header Size = 15 Words (60 Bytes)

4-bit
Unsigned Int
Range
0 to 15



Example 1 :-

[MSQ]



#Q. Which of the following is/are can be a valid IPv4 packet header size in bytes?

☒ A 15

☐ B 20

☒ C 50

☐ D 60

Ans: B & D



Topic : Header Length



Header Size = 5 Words (20 Bytes)

Header Size = 6 Words (24 Bytes)

Header Size = 7 Words (28 Bytes)

Header Size = 8 Words (32 Bytes)

Header Size = 9 Words (36 Bytes)

Header Size = 10 Words (40 Bytes)

Header Size = 11 Words (44 Bytes)

Header Size = 12 Words (48 Bytes)

Header Size = 13 Words (52 Bytes)

Header Size = 14 Words (56 Bytes)

Header Size = 15 Words (60 Bytes)

Example 2 :-

[MSQ]



#Q. Which of the following is/are can be a valid IPv4 packet starting bits?

~~A~~ ^{VER HLEN}
0100 0100

✓ B 0100 0101

✓ C 0100 0110

~~D~~ 0110 0110

Ans: B & C

Example 3 :-

[NAT]



#Q. Consider starting eight bits of an IPv4 packet is "0100 1010", calculate IPv4 packet header size in bytes?

VER HLEN

$$HLEN = (1010)_2 = 10 \text{ words}$$

$$\begin{aligned} \text{Header Size} &= (HLEN * 4) \text{ bytes} \\ &= (10 * 4) \text{ bytes} \\ &= 40 \text{ bytes} \end{aligned}$$

Ans = 40



Topic : Header Length



Size of options (in IPv4 packet header)

$$= [\text{HLEN} - \text{Base Header Size}] \text{ words}$$

$$= [\text{HLEN} - 5] \text{ words}$$

$$= [\text{HLEN} - 5] * 4 \text{ bytes}$$



Topic : Header Length



Maximum Size of options (in IPv4 packet header)

$$= [\text{Maximum Header Size} - \text{Base Header Size}] \text{ words}$$

$$= [15 - 5] \text{ words}$$

$$= 10 \text{ words}$$

$$= 10 * 4 \text{ bytes}$$

$$= 40 \text{ bytes}$$

Example 4 :-

[NAT]



#Q. Consider starting eight bits of an IPv4 packet is "0100 1100", calculate IPv4 packet options size in bytes?

VER HLEN

$$HLEN = (1100)_2 = 12 \text{ words}$$

$$\text{Size of options} = (HLEN - 5) * 4 \text{ bytes}$$

$$= (12 - 5) * 4 \text{ bytes}$$

$$= 28 \text{ bytes}$$

Ans = 28



Topic : Type of Services



- Type of Services [ToS]
- ToS field is 8 bits long
[Second byte of IPv4 packet header]
- For QoS
[Quality of Services]

- ① Delay
- ② Throughput
- ③ Reliability



Topic : Total Length



→ Total Length field is 16 bits long

→ Define size of IPv4 packet (datagram) in bytes (by default)
[including header] ←

→ Maximum IPv4 datagram size = $[2^{16} - 1]$ bytes

$$TL \geq 20$$

Packet or Datagram
= [Header + Payload]



Topic : Total Length



$$\text{Size of Payload} = [\text{Total Length} - (\text{HLEN} * 4)] \text{ bytes}$$

IPv4 Packet



Example 5 :-

[NAT]



#Q. Consider an IPv4 packet, the values in total length field and header length (HLEN) fields are 250 and 10 respectively, calculate size of data (payload size) in bytes carrying by this packet?

Total Length = 250 bytes

HLEN = 10 words

$$\begin{aligned}\text{Payload size} &= [TL - (HLEN \times 4)] \text{ bytes} \\ &= [250 - (10 \times 4)] \text{ bytes} \\ &= 210 \text{ bytes}\end{aligned}$$

Ans = 210

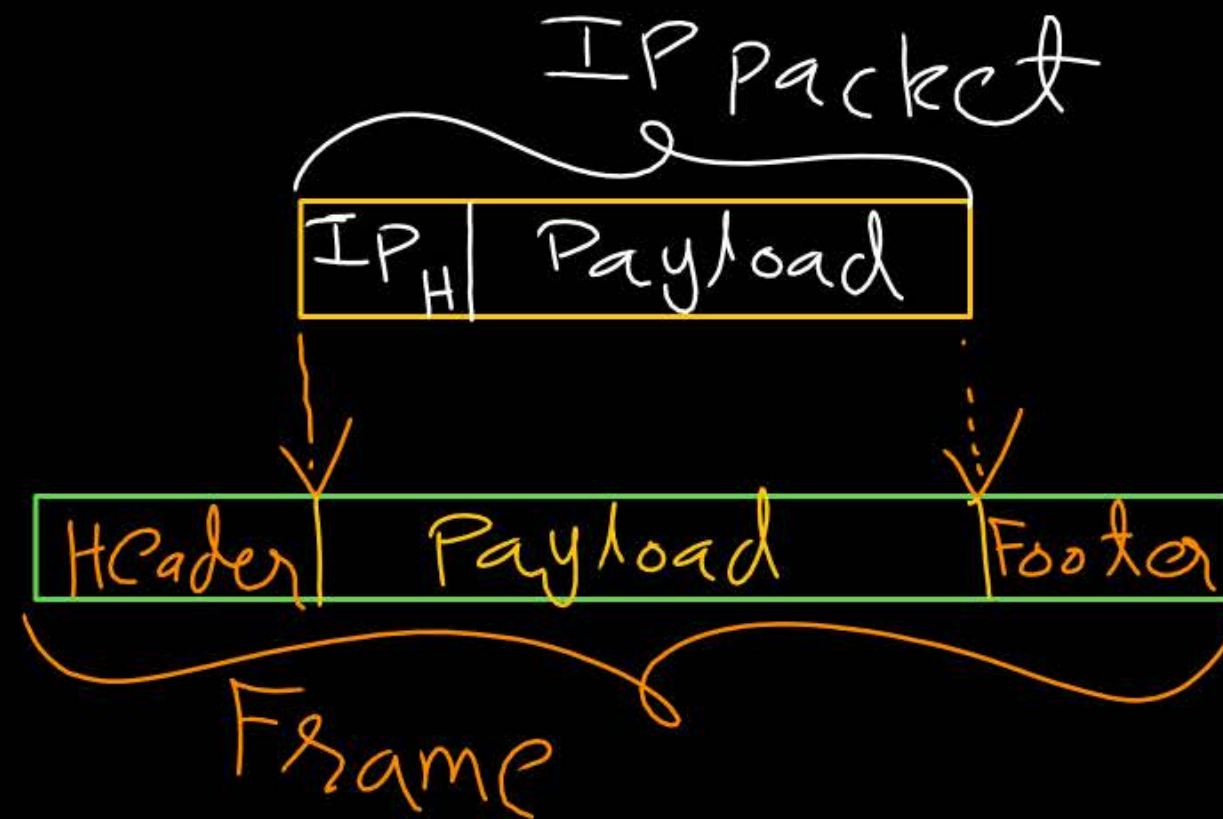


Topic : MTU

$$IP\ Packet\ Size \leq MTU$$

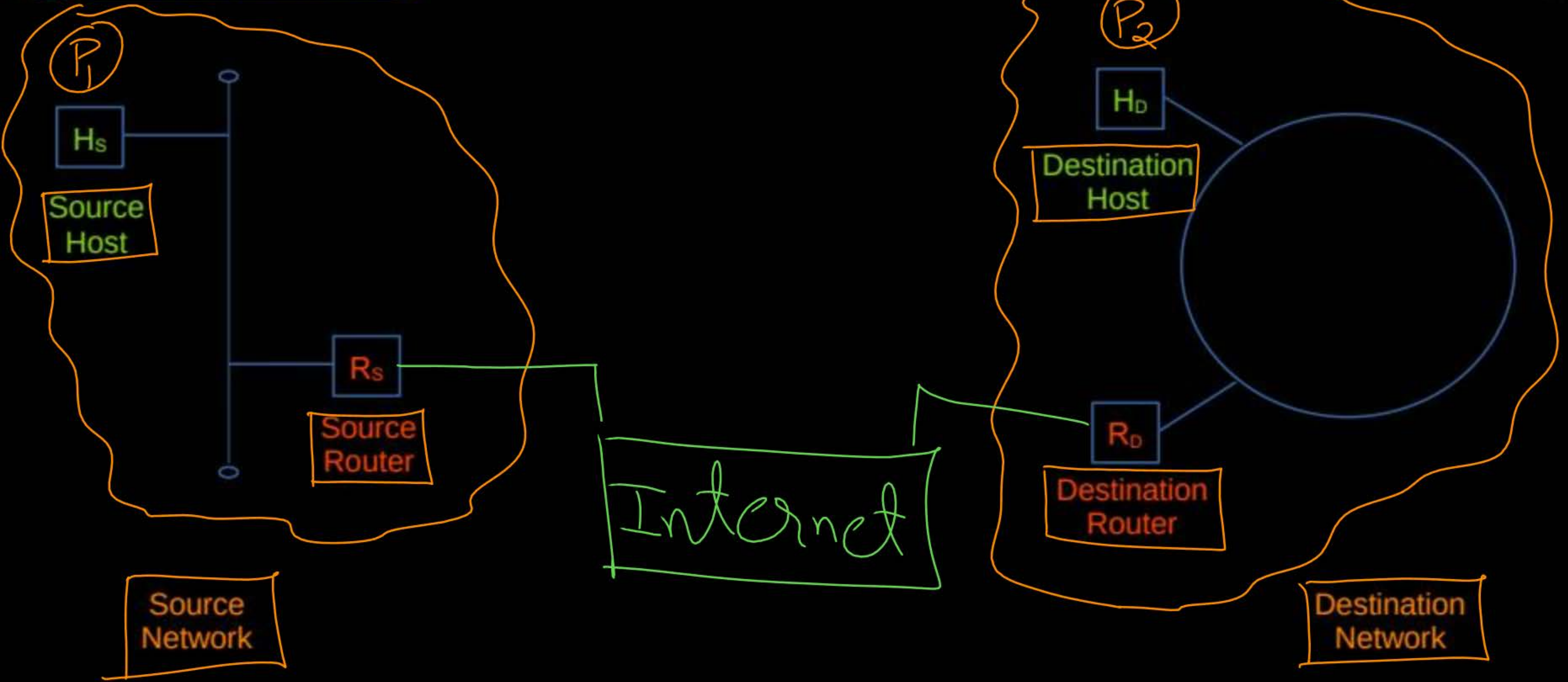


- Maximum Transmission Unit [MTU]
- Measurement in bytes
- Size of largest PDU (IP datagram) that can be communicated over a network





Topic : MTU





Topic : MTU



→ Different networking technologies may have different **MTU**

→ **MTUs** for common media

1. Ethernet v2 :
2. Wi-Fi (**WLAN**, IEEE 802.11) :
3. Token Ring (**IEEE 802.5**) :
4. FDDI :

MTU

1500 ✓

2304

4464

4352

[static value]



Topic : MTU



- Source host creates IPv4 datagram as per source network MTU
- At intermediate IPv4 router, for an received IPv4 datagram
if IPv4 datagram size is greater then next network (link) MTU size
then need to do fragmentation according to MTU



2 mins Summary



Topic

IPv4 Header Format ✓

Topic

IPv4 Header and Datagram size ✓

*
✗

MTU



THANK - YOU

